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IDENTIFICATION OF CONTEXT-INVARIANT PLAYER SKILL METRICS IN ESPORTS THROUGH EFFECT-SIZE ANALYSIS

Yu. I. Kuzhii¹, Yu. M. Furgala¹

¹*Ivan Franko National University of Lviv,
Universytetska Str., 1, Lviv, 79000,
e-mail: yurii.kuzhii@lnu.edu.ua*

ORCID: 0009-0001-7890-8575 (Yu. I. Kuzhii), 0000-0001-7316-7424 (Yu. M. Furgala)

Assessing individual player skill in team-based esports is complicated by the dependence of observed performance metrics on match context — the particular map, the game version, and the side played. This study examines the invariance of 46 Counter-Strike player performance metrics across three dimensions, using 3486 player $\times$ map observations (1820 from Counter-Strike 2, period 2023–2025; 1666 from Counter-Strike: Global Offensive, period 2012–2023) collected from the public resource HLTV.org across eight maps. Invariance is defined by effect size rather than by the p -value, because for samples of several thousand observations the latter indicates significance even for practically negligible differences. One-way ANOVA with η^2 (maps), the Mann–Whitney U test with Cohen’s d (versions), and a paired t -test with Cohen’s d (sides) are applied. We find that 76.1 % of metrics are invariant to the map, 93.5 % to the game version, and only 47.8 % to the side, so that the side is the strongest source of contextual bias. Sixteen metrics are simultaneously invariant across all three dimensions; this “triple-invariant” set (overall rating, firepower, opening-duel and clutch metrics, sniper multi-kills) reflects intrinsic player ability and is suitable for building robust skill-assessment and scouting models. A sensitivity analysis confirms the robustness of the conclusions within each game version: all 33 metrics that are map-invariant in CS2 remain invariant in CSGO as well.

Key words: esports, pattern recognition, feature invariance, effect size, feature selection, skill assessment, analysis of variance.

1. INTRODUCTION

Esports analytics increasingly relies on machine learning and pattern recognition methods to assess individual player skill and to select the features that characterise performance [1]. Counter-Strike is one of the most popular team disciplines, and detailed per-map statistical profiles of its players are publicly available. Observed performance metrics, however, are shaped not only by a player’s individual ability but also by match context: the particular map, the game version (the transition from Counter-Strike: Global Offensive to Counter-Strike 2), and the side played (attack — terrorists, defence — counter-terrorists).

Building robust skill-assessment models requires separating **context-invariant** metrics, which reflect stable player ability, from **context-dependent** ones, which describe the situation rather than the player. This formulation belongs to the classical feature-selection problem [2] and is consistent with the dataset-shift problem, in which the properties of the feature space change between observation conditions [3].

The aim of this work is to identify Counter-Strike player performance metrics that are invariant with respect to the map, the game version, and the side, and to isolate the subset that is invariant across all three dimensions simultaneously. Unlike the common approach based on statistical significance, we define invariance through **effect size**, because at a sample size of several thousand observations even practically negligible differences become statistically significant [5, 6].

2. MATERIALS AND METHODS

2.1. DATASET

The data were collected from the public resource HLTV.org — per-map statistical profiles of players. Each observation corresponds to a player $\times$ map pair and contains an aggregated profile of metrics overall and separately for the CT and T sides. After merging the two game versions, filtering valid maps (names with the `de_` prefix) and discarding records without a version label, **3486 observations** were obtained (CS2 — 1820, period 2023-09-27... 2025-12-31; CSGO — 1666, period 2012-08-21... 2023-09-26) for **8 maps**: `de_ancient`, `de_anubis`, `de_dust2`, `de_inferno`, `de_mirage`, `de_nuke`, `de_train`, `de_vertigo`.

Forty-six baseline (overall) performance metrics were analysed, corresponding to the seven play-style attributes of the HLTV taxonomy [7]: firepower, entrying, trading, opening duels, clutching, sniping, and utility usage. All numeric columns were cast to a numeric type with correct handling of missing-value markers.

2.2. DEFINITION OF INVARIANCE

A metric is considered **invariant** with respect to a dimension if the effect size of the between-group differences does not exceed the moderate threshold of Cohen’s classification [5]. p -values are reported for reference but are not used as a criterion, because at $n \approx 3486$ they almost always indicate significance regardless of the practical magnitude of the differences [5, 6].

2.3. STATISTICAL METHODS

Maps (8 groups, independent). One-way analysis of variance (ANOVA) with effect size $\eta^2 = SS_{\text{between}}/SS_{\text{total}}$. Invariance threshold: $\eta^2 < 0.06$ (below a medium effect).

Game version (2 groups, independent). The non-parametric Mann–Whitney U test (robust to the departures from normality and to the outliers typical of game statistics) with effect size Cohen’s $d = |\mu_1 - \mu_2|/\sigma_{\text{pooled}}$, where $\sigma_{\text{pooled}} = \sqrt{(\sigma_1^2 + \sigma_2^2)/2}$. Threshold: $d < 0.5$. Note that the test is rank-based whereas Cohen’s d is a parametric measure; we combine them deliberately in order to keep a single effect-size scale across all three dimensions, while acknowledging that a rank-based measure (Cliff’s δ , for instance) would be strictly more consistent.

Side (2 groups, paired). Every player has paired CT and T observations within the same profile, so a paired t -test was applied with the paired Cohen’s $d = |\mu_{\text{diff}}|/\sigma_{\text{diff}}$ (the within-subject form). The threshold values are borrowed from Cohen’s conventions for independent samples; for the paired d this makes the effect-size estimates somewhat more conservative, which should be taken into account when interpreting the side dimension specifically.

Handling of missing values. Each test uses the observations available for the corresponding metric (pairwise deletion), so the sample size depends on the metric: for most metrics the share of missing values did not exceed 1 %, reaching 9–21 % only for derived sniper percentage metrics (for example, `1on1_win_percentage` has $n_{\text{CS2}} = 1754$; for the paired side analysis `1on1_win_percentage` yields 3026 pairs against 3486 for complete metrics). All n values are recorded in the output tables.

Dimensionality reduction. Principal component analysis (PCA) on the standardised overall metrics, used to visualise the separability of groups by map and by version [4].

Intersections. The “universally invariant” set is the intersection of the map- and version-invariant metrics; the “triple-invariant” set adds the intersection with the side-invariant ones.

Sensitivity analysis. Because the game version is a potential confounder in the map analysis (the data are pooled), map invariance was additionally computed separately within each version.

All computations were carried out by reproducible scripts (Python: pandas, numpy, scipy, scikit-learn, matplotlib); the output tables and figures are generated automatically.

3. RESULTS

3.1. INVARIANCE TO THE MAP

Of the 46 metrics, **35 (76.1 %)** are invariant to the map ($\eta^2 < 0.06$). Invariance is highest for the opening-duel metrics (`opening_deaths_per_round`, `opening_attempts`) and for 1-on-1 duels (`1on1_win_percentage`), all with $\eta^2 < 0.001$. The strongest map dependence is found in utility-related metrics: `utility` ($\eta^2 = 0.247$), `utility_damage_per_round` ($\eta^2 = 0.243$), `flashes_thrown_per_round` ($\eta^2 = 0.209$). The distribution of effect sizes across all three dimensions is shown in Fig. 1.

3.2. INVARIANCE TO THE GAME VERSION (CS2 VS CSGO)

Forty-three of the 46 metrics (93.5 %) are invariant to the version (Cohen’s $d < 0.5$). The most stable are `opening_deaths_per_round` ($d = 0.003$), `utility` ($d = 0.004$), `last_alive_percentage` ($d = 0.015$). The largest differences appear in team-interaction metrics: `assisted_kills_percentage` ($d = 2.51$; 27.1 % in CS2 against 17.8 % in CSGO), `assists_per_round` ($d = 1.80$), `support_rounds` ($d = 0.96$). The distributions of key features for the two versions are shown in Fig. 2.

3.3. INVARIANCE TO THE SIDE (T VS CT)

The side has the strongest influence: only **22 of the 46 metrics (47.8 %)** are invariant. The most stable are `opening_attempts` ($d = 0.001$), `entrying` ($d = 0.005$), `pistol_round_rating` ($d = 0.009$). The most side-dependent are `saves_per_round_loss` ($d = 1.96$), `trade_kills_percentage` ($d = 1.00$), `saved_by_teammate_per_round` ($d = 0.98$), `flashes_thrown_per_round` ($d = 0.97$). A comparison of the mean values by side is given in Fig. 3.

3.4. TRIPLE-INVARIANT METRICS

The intersection of the three dimensions yields **33 universally invariant** metrics (map + version) and **16 triple-invariant** ones (map + version + side), listed in Table 1.

3.5. PCA AND SENSITIVITY ANALYSIS

The first two principal components explain 23.1 % and 18.2 % of the variance (41.3 % together). In the PCA projection the groups by map and by version overlap substantially (Fig. 4), which visually confirms the predominant invariance of the feature space. The sensitivity analysis produced similar shares of map-invariant metrics within each version — 71.7 % (CS2, 33/46) and 76.1 % (CSGO, 35/46) against 76.1 % on the pooled data. Importantly, these shares are a quantity distinct from the share of the universally invariant set (also 33/46), with which they coincide numerically. The sets are stable in composition: all 33 metrics that are map-invariant in CS2 are also invariant in CSGO (the difference

Feature invariance by effect size (green = invariant, red = variant)



Figure 1. Effect sizes for the three invariance dimensions (the 10 most and the 10 least invariant features per panel). Green — invariant ($\eta^2 < 0.06$ for maps; Cohen's $d < 0.5$ for version and side), red — variant. Dashed line — threshold

Table 1. Triple-invariant metrics (invariant to the map, the version and the side)

<i>Category (HLTV attribute)</i>	<i>Metrics</i>
Firepower	firepower; damage_per_kill
Entrying	entrying; traded_deaths_percentage; opening_deaths_traded_percentage
Opening duels	opening; opening_attempts; win_after_opening_kill
Clutching	clutching; clutch_points_per_round; 1on1_win_percentage
Sniping	sniping; sniper_multi_kill_rounds; sniper_opening_kills_per_round
Overall / situational (outside the seven HLTV attributes)	rating_3.0; pistol_round_rating

amounts to two additional metrics in CSGO: `assisted_kills_percentage`, `attacks_per_round`), so pooling the versions does not distort the conclusions about maps.

4. DISCUSSION

The results outline a clear hierarchy of contextual influence: the weakest is the game version (7 % of variant metrics), moderate is the map (24 %), and the strongest is the side (52 %). This means that for cross-context skill assessment the side is the decisive source of bias, and metrics should either be normalised by side or restricted to the side-invariant set.

The pattern of results is consistent with game logic. The strong map dependence of utility usage is explained by the different grenade strategies that different maps require, whereas the stability of the opening-duel metrics reflects individual skill in the first engagements. The predominant invariance to the version indicates that basic mechanical skills transfer between CSGO and CS2, while the largest differences are concentrated in assist and support metrics, which may be related to the mechanical changes introduced in CS2; this statement is correlational and requires separate verification. The dependence of save and utility metrics on the side matches the different economic and positional tasks of CT and T.

The triple-invariant metrics mostly concern individual aiming and decision-making skills in key situations (first engagements, clutches, pistol rounds) rather than team coordination (assists, support, saves), which is expected to depend on role and side. This makes them natural candidates for features in skill-assessment and scouting models.

Compared with the common approach based on statistical significance, defining invariance by effect size yields substantively different results: at $n \approx 3486$ the overwhelming majority of metrics differ “significantly” ($p < 0.05$) even for negligible differences, so the significance criterion is not applicable here.

Limitations. (1) The data are aggregated at the map level, without distinguishing opponents or a player’s period of form. (2) The CS2 period is shorter than the CSGO one, which may induce a shift in the player pool. (3) Some sniper percentage metrics have up to 21 % missing values. (4) The analysis is correlational; causal statements about game mechanics require additional verification. (5) Confidence intervals for η^2 and Cohen’s d were not computed; adding them (by bootstrap, for instance) would strengthen the central argument about effect sizes.

Distributions of key features: CS2 vs CSGO

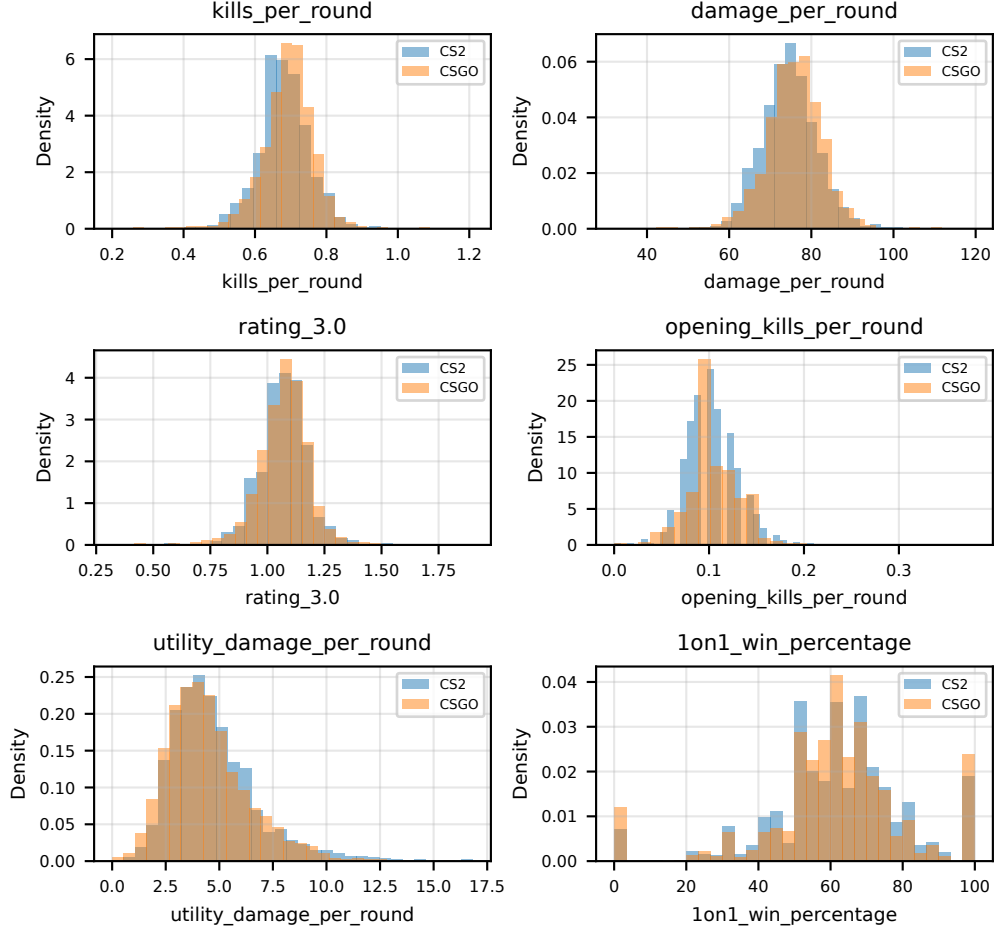


Figure 2. Distributions (density) of key features for CS2 and CSGO

5. CONCLUSIONS

1. Context-invariant performance metrics of Counter-Strike players were identified on a set of 3486 observations across three dimensions on the basis of effect sizes.
2. Invariance holds for 76.1% of the metrics with respect to the map, 93.5% with respect to the game version, and 47.8% with respect to the side; the side is the strongest source of contextual bias.
3. Sixteen triple-invariant metrics were isolated; they reflect intrinsic player ability and are suitable for robust feature selection.
4. The advantage of the effect-size criterion over statistical significance was demonstrated for large samples; the sensitivity analysis confirmed the robustness of the conclu-

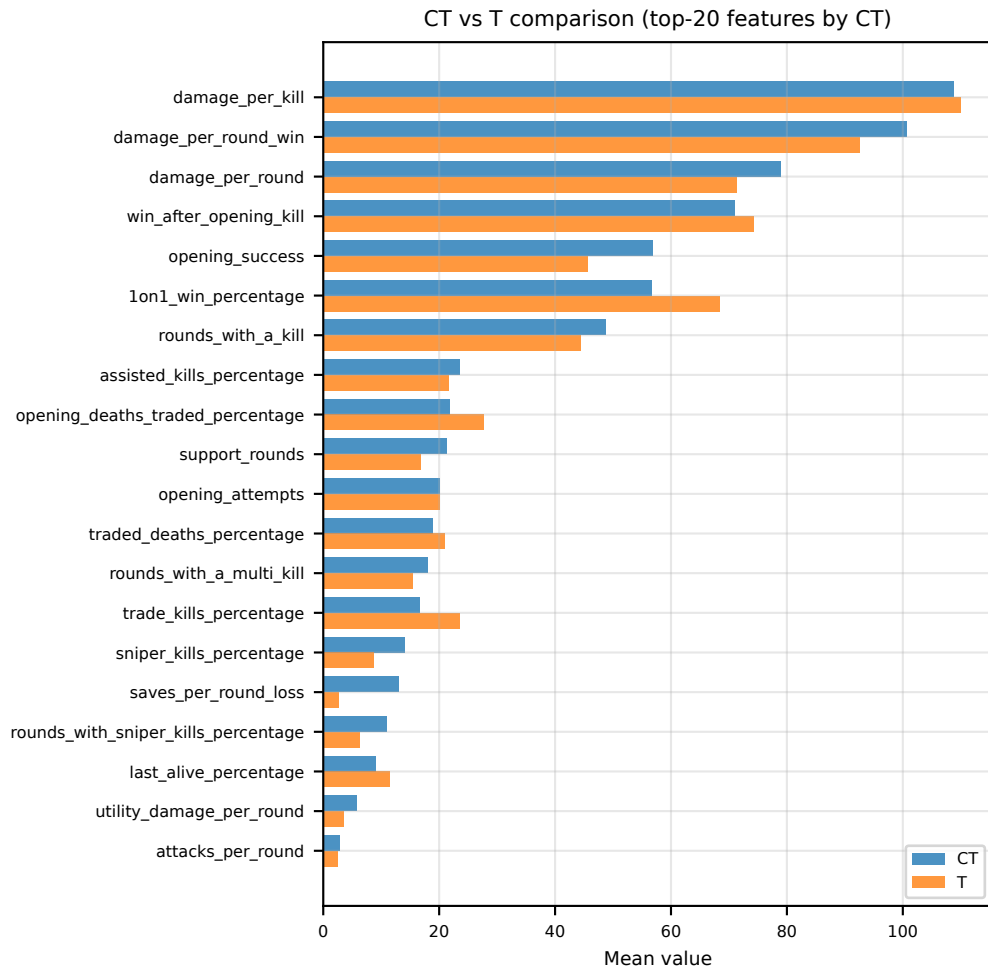


Figure 3. Mean feature values for the CT and T sides (top-20 features by CT). The largest gaps are in save and utility metrics; opening and clutch metrics nearly coincide

sions within each game version.

5. A prospective direction is the application of the triple-invariant set in skill-assessment models and in the detection of anomalous player behaviour patterns, as the next stage of the publication series.

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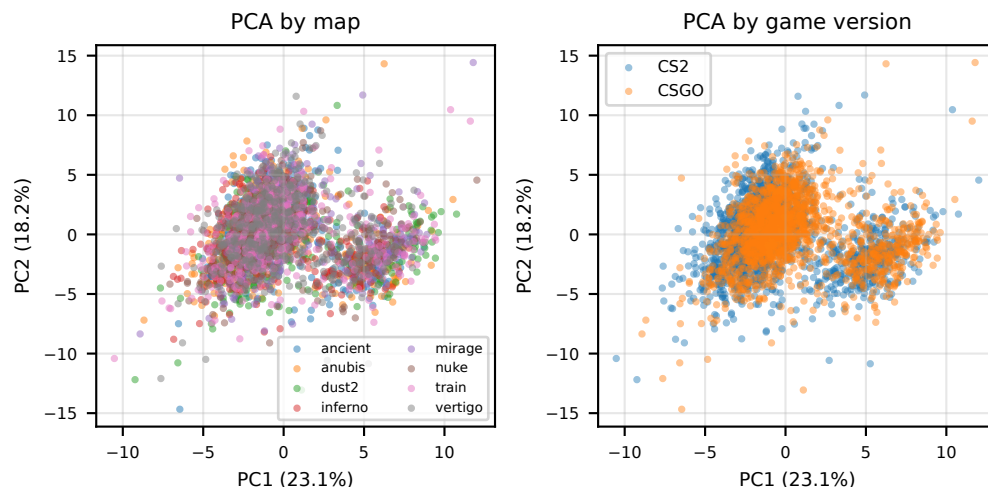


Figure 4. PCA projection of the standardised overall feature space onto two principal components (PC1 — 23.1%, PC2 — 18.2%). Left — by map, right — by game version

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ІДЕНТИФІКАЦІЯ КОНТЕКСТНО-ІНВАРІАНТНИХ МЕТРИК МАЙСТЕРНОСТІ ГРАВЦЯ У КІБЕРСПОРТІ НА ОСНОВІ АНАЛІЗУ РОЗМІРІВ ЕФЕКТУ

Ю. І. Кузій¹, Ю. М. Фургала¹

¹Львівський національний університет імені Івана Франка,
вул. Університетська, 1, Львів, 79000,
e-mail: yurii.kuzhii@lnu.edu.ua

ORCID: 0009-0001-7890-8575 (Ю. І. Кузій), 0000-0001-7316-7424 (Ю. М. Фургала)

Оцінювання індивідуальної майстерності гравця у командних кіберспортивних дисциплінах ускладнюється тим, що спостережувані метрики продуктивності залежать від контексту матчу — конкретної карти, версії гри та сторони. У роботі досліджено інваріантність 46 метрик продуктивності гравців Counter-Strike за трьома вимірами на наборі з 3486 спостережень «гравець × карта» (1820 — Counter-Strike 2, період 2023–2025; 1666 — Counter-Strike: Global Offensive, період 2012–2023), зібраному з відкритого ресурсу HLTV.org для восьми карт. Інваріантність визначено за розміром ефекту, а не за p -значенням, оскільки за вибірок у кілька тисяч спостережень останнє вказує на значущість навіть за практично незначних відмінностей. Застосовано однофакторний дисперсійний аналіз з η^2 (карти), критерій Манна-Уїтні з Cohen's d (версії) та парний t -критерій з Cohen's d (сторони). Встановлено, що 76,1% метрик інваріантні до карти,

93,5 % — до версії гри і лише 47,8 % — до сторони, тобто сторона є найсильнішим джерелом контекстного зміщення. Шістнадцять метрик виявилися інваріантними одночасно за всіма трьома вимірами; цей «триінваріантний» набір (загальний рейтинг, вогнева міць, метрики відкриваючих дуелей і клатчів, снайперські мультікіли) відображає внутрішні здібності гравця і придатний для побудови робастних моделей оцінювання майстерності та скаутингу. Аналіз чутливості підтвердив стійкість висновків у межах кожної версії гри: усі 33 пар-інваріантні в CS2 метрики залишаються інваріантними і в CSGO.

Ключові слова: кіберспорт, розпізнавання образів, інваріантність ознак, розмір ефекту, відбір ознак, оцінювання майстерності, дисперсійний аналіз.